

publicly visible development set predictions on the CodaLab platform. We removed six submissions whose exact match (EM) scores on test data were less than 50%; EM scores below 50% suggest a bug or misconfiguration of the model for predicting on the test set, as the majority baseline gets roughly 50% accuracy (by always predicting no-answer). We also removed one submission whose development set EM score was more than 20 points higher than its test EM score, as it seemed likely that the model had been trained on the development set. After this filtering, we were left with 144 models.

Figure 9 shows the correlation between validation and test data for both pairwise accuracy differences (Δ_{acc}) and agreement rates (P_a) on the SQuAD 2.0 leaderboard. As can be seen, these correlate well, suggesting that measuring these quantities on validation data can serve as a reasonable guide when doing a power analysis for a new model, though lower agreement rates on dev data to tend to slightly underestimate agreement on test. If the validation results are available for both models, these can be used to compute estimates of P_a and Δ_{acc} , and these can be used to compute the approximate power of the test set.

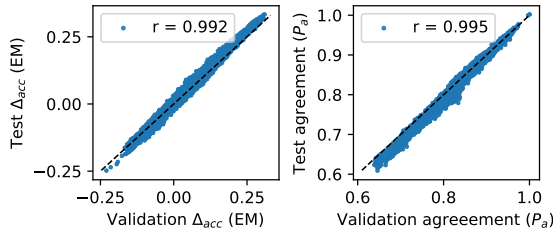


Figure 9: Correlation between validation and test data among all models submitted to the SQuAD 2.0 leaderboard for both pairwise accuracy differences (Δ_{acc} using exact match (EM); left), and agreement rates (P_a ; right). In both cases, Pearson correlation (r) is over 0.99. Dashed lines show $y = x$.

To verify that using these estimates provide a reliable guide to power, we make use the predictions made by SQuAD 2.0 submissions on both validation and test data. In particular, if we assume that each submission is being compared to the previous model to demonstrate a significant and well-powered improvement over the previous baseline, we find that 19 out of 143 submissions showed sufficient improvement on the validation set to have at least 80% power (see Figure 10). Of these, 14 (74%) attain a significant improvement over the baseline on the test data (consistent with

the expected value of 80%). Of the remaining 124 submissions, 3 (2.5%) would show a significant improvement over the baseline, but did not have sufficient power based on validation performance. Interestingly, while all other significant improvements were generally well-spaced over time, these three underpowered submissions were all beaten by a new submissions within 5 days. As an aside, we also note that the vast majority of submissions are significantly worse than the current SOTA, reinforcing the notion that real improvements are rare, and most improvements will be small.

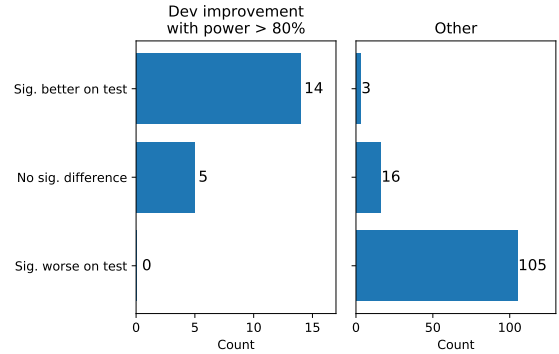


Figure 10: SQuAD 2.0 leaderboard submissions compared to previous SOTA, where we require for SOTA that submissions have 80% power (based on validation improvement and agreement), and a significant improvement on test data.

Caveats: Correlation between the effect size on the validation and test sets may not always be so high. Overconfidence in the power of your experiment may thus occur if the validation performance is greater than the test performance (as would be the case if no regularization was used and extensive hyperparameter tuning caused a model to overfit to the validation set). Alternatively, if comparing to a baseline with inflated performance on validation data (for the same reasons as above), running power analyses based purely on estimates from validation data would underestimate power. As such, combining validation estimates with reasonable priors is recommended.

E Accuracy

E.1 Data Collection

E.1.1 Model Predictions on Test Set and Model Prediction Agreement

From the authors of the GLUE benchmark – as well as authors of individual models – we obtain

the model test-set predictions on all tasks from a set of 10 high-performing models, which allows us to measure the extent to which their predictions overlap with each other. We select GLUE tasks which use accuracy as an evaluation metric. The relevant tasks are MNLI (Williams et al., 2018), MRPC (Dolan and Brockett, 2005), RTE (Dagan et al., 2005; Bar-Haim et al., 2006; Giampiccolo et al., 2007; Bentivogli et al., 2009), SST-2 (Socher et al., 2013), QQP (Iyer et al., 2017), QNLI (Rajpurkar et al., 2016), and WNLI (Levesque et al., 2012). For consideration of other metrics, see Appendix F.

We use model predictions for: ELECTRA (small, base, large, large with tricks) (Clark et al., 2019b), XLNet (large) (Yang et al., 2019), T5 (Raffel et al., 2019), ALBERT (large) (Lan et al., 2020), BAM(large) (Clark et al., 2019a), RoBERTa (large) (Liu et al., 2019), and BERT (Devlin et al., 2019). We only had the model predictions available and extrapolated overlap from that, we did not have access to the models themselves, ground truth test set labels, nor dev set predictions for the models.

E.1.2 Comparisons and Claims

We gather data from GLUE papers regarding the accuracy tasks and manually label 119 comparisons and 57 claims of improvement (as denoted within a work by bolding of a new model’s number and a claim of SOTA in the main text) across 14 papers (selected as being at or above the BERT score on the GLUE leaderboard with an accompanying publication). For each paper we examine if a specific comparison is made against a baseline that isn’t claiming state of the art performance. For example, the STILTs approach (Phang et al., 2018) makes comparisons against non-SOTA baselines, which we add to our labeling scheme but filter out when fitting regressions to likely SOTA improvements. We mark this as **SOTA Comparison** = N. For claims of SOTA improvement, we examine this as some textual basis for the claim (e.g., “we drive state of the art performance on GLUE”) coupled with bolding of values in a table reporting baselines against the model under test. We mark datapoints as **Claim of Improvement** = Y if they are an improvement claim. We mark effect size as the improvement from the best previous baseline (the current SOTA) on the test set on a per-dataset basis. We note that in several cases, worse results on the new model were bolded. We treated this as no claim of improvement. If results were not

bolded but still higher for the new model we also treated this as no claim for improvement.

E.2 Regression-based approach to modeling power and MDEs

E.2.1 Predicting overlap

There are several versions of McNemar’s test, each with their own unique method for calculating power, sample size, or minimum effect size. See, for example, discussions in Schlesselman (1982), Duffy (1984) Suissa and Shuster (1991), Connett et al. (1987), Fagerland et al. (2013), and Lachenbruch (1992).

The methods for calculating sample size or power by Connett et al. (1987); Schlesselman (1982); Suissa and Shuster (1991) require making an assumption about the odds ratio $\Phi = p_{10}/p_{01}$ as well as an estimate of the fraction of discordant pairs (disagreements between two models).

Fagerland et al. (2013) suggest that the exact unconditional version of the test by Suissa and Shuster (1991) has desirable properties. Thus, we use the implementation of the power calculations for this test from the <https://github.com/ekstroem/MESS> package.

How do we make an assumption about the odds ratio and fraction of discordant pairs? We first fit an OLS regression to the existing models on the GLUE leaderboard for all binary choice accuracy tasks using the aforementioned predictions provided by the leaderboard creators and individual authors of models,

$$\text{overlap}_i = \beta_0 + \beta_1 \text{min_acc}_i + \beta_2 \text{acc_diff}_i, \quad (1)$$

for all i that are a pairwise comparison between any two models, min_acc_i is the minimum accuracy between the two models under comparison, acc_diff_i is the gap between the two models, and overlap_i is the fraction of overlapping predictions. We end up with the model shown in Table 5.

We note that outcomes are biased toward a higher range of accuracy values and may not be a perfect prior. However, this does give us a fairly good linear fit for top-of-the-leaderboard results. We then can predict the expected overlap for a given model as:

$$\begin{aligned} \text{exp_overlap} = & 0.41 + 0.58 \cdot \text{min_acc} \\ & - 0.47 \cdot \text{exp_acc_dif} \end{aligned} \quad (2)$$

Note now we can make an assumption on the expected fraction of discordant values and the odds

Dep. Variable:	y	R-squared:	0.966
Model:	OLS	Adj. R-squared:	0.966
Method:	Least Squares	F-statistic:	3820.
Date:	Thu, 14 May 2020	Prob (F-statistic):	3.62e-197
Time:	07:03:28	Log-Likelihood:	818.14
No. Observations:	270	AIC:	-1630.
Df Residuals:	267	BIC:	-1619.
Df Model:	2		

	coef	std err	t	P> t	[0.025	0.975]
const	0.4142	0.019	21.694	0.000	0.377	0.452
min_acc	0.5819	0.021	27.999	0.000	0.541	0.623
acc_diff	-0.4662	0.028	-16.625	0.000	-0.521	-0.411

Omnibus:	6.121	Durbin-Watson:	1.040
Prob(Omnibus):	0.047	Jarque-Bera (JB):	8.647
Skew:	-0.108	Prob(JB):	0.0133
Kurtosis:	3.850	Cond. No.	71.5

Table 5: OLS Regression Results for predicting GLUE model overlap from baseline accuracy and effect size.

Dep. Variable:	y	R-squared:	0.944
Model:	OLS	Adj. R-squared:	0.933
Method:	Least Squares	F-statistic:	91.87
Date:	Tue, 26 May 2020	Prob (F-statistic):	1.37e-07
Time:	06:05:23	Log-Likelihood:	36.368
No. Observations:	14	AIC:	-66.74
Df Residuals:	11	BIC:	-64.82
Df Model:	2		

	coef	std err	t	P> t	[0.025	0.975]
const	0.4339	0.091	4.786	0.001	0.234	0.633
min_acc	0.5932	0.101	5.874	0.000	0.371	0.816
acc_diff	-1.2849	0.588	-2.186	0.051	-2.578	0.009

Omnibus:	0.299	Durbin-Watson:	2.022
Prob(Omnibus):	0.861	Jarque-Bera (JB):	0.163
Skew:	0.214	Prob(JB):	0.922
Kurtosis:	2.691	Cond. No.	140.

Table 6: OLS Regression Results for predicting SQuAD 2.0 model overlap.